

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/232131414>

Guiding Coaches Through Scientific Articles by Examining Human Growth Hormone Research

Article in *Strength and conditioning journal* · October 2011

DOI: 10.1519/SSC.0b013e31822f3421

CITATIONS

3

READS

727

2 authors:



Andrew J Galpin

California State University, Fullerton

164 PUBLICATIONS 491 CITATIONS

[SEE PROFILE](#)



James R Bagley

San Francisco State University

77 PUBLICATIONS 176 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Muscle Fiber Adaptations to Simulated Microgravity [View project](#)



Virtual Reality Health and Exercise [View project](#)

Guiding Coaches Through Scientific Articles by Examining Human Growth Hormone Research

Andrew J. Galpin, PhD, CSCS, NCSA-CPT¹ and James R. Bagley, MS²

¹Center for Sport Performance, Department of Kinesiology, California State University, Fullerton, California; and

²Human Performance Laboratory, Ball State University, Muncie, Indiana

SUMMARY

COACHES AND PRACTITIONERS RARELY RECEIVE EXTENSIVE TRAINING IN IDENTIFYING AND INTERPRETING PEER-REVIEWED RESEARCH, MAKING IT NEARLY IMPOSSIBLE FOR THEM TO ACCURATELY DETERMINE THE VALIDITY OF POPULAR TRAINING, EQUIPMENT, AND SUPPLEMENT PRODUCTS. THEREFORE, THE PURPOSE OF THIS PAPER IS TO (A) DESCRIBE A QUICK AND EASY PROCESS (RAPID METHOD) FOR COACHES TO GATHER INFORMATION AND MAKE EDUCATED DECISIONS ON UNFAMILIAR TOPICS AND (B) APPLY THIS METHOD TO A POPULAR ISSUE: HUMAN GROWTH HORMONE SUPPLEMENTATION.

INTRODUCTION

Coaches are often expected to answer questions regarding unfamiliar topics such as new training fads, products, or supplements. Thus, it is important to keep up with popular claims and scientific research by continually gathering knowledge from external sources. Yet finding quality, scientific evidence on the information-

oversaturated Internet is tremendously difficult and time consuming. Realizing coaches are extremely busy, we present a streamlined process (termed the RAPID method, Figure 1) to help them find, analyze, and interpret peer-reviewed research. This 5-step method will be applied to the topic of human growth hormone (HGH) supplementation.

Illegal use of HGH has exploded in recent years as professional organizations (Major League Baseball, National football League, International Olympic Committee, etc) scramble to develop abuse detection methods. This sudden increase in HGH popularity can be traced back (5) to a landmark *New England Journal of Medicine* study by Rudman et al. (4) in 1990 on therapeutic HGH supplementation in elderly humans. Yet rifts exist between the proposed and scientifically documented value of HGH as a performance-enhancing drug in young/healthy individuals (1–3). Therefore, the purpose of this article is to introduce the RAPID method by applying it to HGH supplementation research, specifically highlighting the article by Rudman et al. (4).

RAPID METHOD

STEP 1: RECOGNIZE THE QUESTION

When presented with a question from one of your athletes, make sure you clearly recognize what is being asked. Once clarified, write down the question concisely in a journal, notebook, or computer file.

HGH example: *What are the scientifically documented effects/benefits of HGH supplementation?*

STEP 2: ASSESS POPULAR CLAIMS

Use second-hand sources, such as search engines, blogs/forums, and professional opinions, to gather key words and basic subject knowledge, but do not form conclusions at this point. Even if second-hand sources cite specific laboratory studies, do not assume that the claims are accurately referenced.

HGH example: *Claims that HGH positively impacts sleep quality, vitality, immune system function, wound healing, and vision prompted the outburst of an entire anti-aging marketplace, calling it*

KEY WORDS:

performance-enhancing drugs; human skeletal muscle; reading research

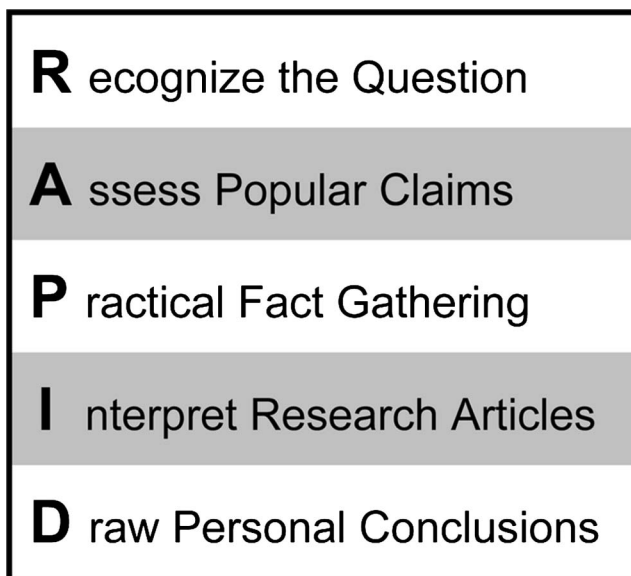


Figure 1. RAPID method of finding, analyzing, and interpreting scientific research.

a “fountain of youth” drug. Moreover, assertions that HGH increases skeletal muscle size and strength likely tempt many athletes to use it illegally.

STEP 3: PRACTICAL FACT GATHERING

Practical fact gathering requires distinguishing between original research (peer-reviewed, laboratory studies) and second-hand research (opinions from websites, blogs, books, etc). Use original, not second-hand, research as the foundation for your opinions because these articles must undergo expert inspection before publication (peer-review process). PubMed (<http://www.ncbi.nlm.nih.gov/pubmed/>) and Google Scholar (<http://scholar.google.com/>) are 2 popular online sites that provide original research articles. Sometimes they offer entire articles but minimally give the title and abstract free of charge. University and institution library websites provide most articles free for affiliates (i.e., students, staff, and employees), and the National Strength and Conditioning Association (NSCA) or other scientific/professional organizations provide members online access to their publications. Proper use of search engine key words helps tremendously when filtering through mountains of research. A few dozen articles can quickly be sifted

through by looking at the titles and abstracts (paper summary, typically <250 words). However, do not make conclusions from Titles and Abstracts because they are sometimes misleading and do not always include all pertinent information.

Original research articles generally come in 2 forms, review articles (synthesis of all pertinent studies) and single scientific papers (detailed descriptions of 1 study). Read review articles cautiously because they provide only one author’s synopsis on the subject. Yet, their reference sections (also known as a bibliography or works cited) can be used to dramatically improve search speed because they supply in-depth lists of relevant single papers. Develop your opinion only after obtaining several single papers, given that 1 paper does not “prove” anything, but many articles sharing similar conclusions build a strong case.

HGH example: A PubMed search using the key terms “human growth hormone” produces >30,000 articles. This can be narrowed to <600 by increasing specificity of the search to “human growth hormone skeletal muscle.” Adding the word *hypertrophy* drops this to a manageable 34 (24 original investigations and 10 review articles). A diligent assessment of all 34

would be ideal, yet probably not realistic; so we encourage starting with the review articles. The titles indicate 5 are of molecular and structural concerns, 3 are clinically based (HGH and renal/heart failure, etc), and only 2 address HGH supplementation and human skeletal muscle performance. The latter 2 articles (3,6) should be read thoroughly because they are short, well written, and easy to follow (even for nonscientists). These review articles and several second-hand sources appear to be greatly influenced by the aforementioned article by Rudman et al. (4). Therefore, in the following 2 steps, we will analyze and interpret this freely available article (<http://www.nejm.org/doi/pdf/>).

STEP 4: INTERPRET RESEARCH ARTICLES

Typical peer-reviewed articles follow a standard format: introduction, methods, results, and discussion.

Reading the Introduction. Introductions provide background information but are not complete reviews of literature; they encompass a few hand-selected references that sell the study’s importance to readers and reviewers. Although this is a good place to find other pertinent references, do not draw conclusions here. Focus on the purpose of the study, usually stated clearly in the first and/or last paragraph.

HGH example (Rudman, 1990): *The authors hypothesize that because both HGH concentrations and body composition decline with age, HGH replacement therapy may attenuate the loss of lean body mass. This was supported by past studies, in which lean body mass improved after HGH treatment in previously HGH-deficient rodents, children, women, and men (age, 20–50 years). Therefore, the purpose of the study was to assess the influence of 6 months of HGH supplementation on markers of body composition and bone density in HGH-deficient men aged 61–81 years.*

Evaluating the Methods. When reading this section, focus primarily on the subject population (species, age, gender, healthy or diseased, training status, etc), measured variables that may/may not change (strength, muscles size,

Table 1

Recommended sample log sheet for applying the RAPID method to an original research article (Rudman et al. (4))

Article Title: Effects of Human Growth Hormone in Men Over 60 Years Old
Author: Rudman, D.
Journal: The New England Journal of Medicine
Date: July 5, 1990
Purpose: Determine the influence of 6 months of HGH replacement therapy on markers of body composition in 61- to 81-year-old HGH-deficient men.
Subjects: Twenty-one healthy, 61- to 81-year-old HGH-deficient men.
Measured Variables: Weight, LBM, adipose tissue, skin thickness, bone density (arm, lumbar spine, hip, and thigh), and jaw height.
Interventions: Six months of physician-monitored, dosage-adjusted HGH replacement therapy injections.
Primary Findings: Weight ↔, LBM ↑, adipose tissue ↓, skin thickness ↑, bone density (arm ↔, lumbar spine ↑, hip ↔, and thigh ↔), and jaw height ↔.
Your Conclusion: 2% ↓ in adipose tissue and 4% ↑ in LBM is intriguing but need more robust measures. <1% ↑ in skin thickness and lumbar density are less impressive. All other markers were not responsive. Overall, appears to have little influence on bone density but does influence fat composition. However, these men were 61- to 81-years old and HGH deficient; this just brought them back to normal values. Also, HGH was given using physician injections and HGH levels were measured constantly. This enabled the physician to adjust dosages and therefore maximize efficacy. This is critical and not possible with people using HGH illegally.
General Comments: Conclusions stated in the abstract are inappropriate. The study design does not allow for a “cause and effect” conclusion: Did they get stronger? More powerful? Nice job of describing study limitations (i.e., only applicable to 61- to 81-year-old HGH-deficient men).
↑ = increase, ↓ = decrease, and ↔ = no change.

hormone concentrations, performance, etc), and interventions (training, feeding, drug therapy, etc). Nonresearchers should spend minimal time reading detailed lab methodologies. Instead, take note of how measurements were made to allow comparisons between other similar studies.

HGH example (Rudman, 1990): *Subjects—Twenty-one HGH-deficient, but otherwise healthy, men (age, 61–81 years) were divided into an experimental group (12 men) and a control group (9 men). Measured Variables—Pretreatment and post-treatment weight, lean body mass, adipose tissue mass (all by hydrostatic weighing), skin thickness, and bone density. Intervention—The control group continued their normal lives, whereas the experimental group received injections of HGH 3 times per week for 6 months. The dosage was adjusted each month to maintain optimal body concentrations. All men were told to resume normal daily activity, diet, and lifestyle.*

Interpreting the Results. This section is the most important, so look carefully at tables and figures as they illustrate the study’s major findings. Although a study may show an increase or decrease in a given measurement, it should not be considered significant unless determined statistically. Before proceeding to the Discussion, highlight and consider pertinent results in context with the article’s purpose (found in the Introduction).

HGH example (Rudman, 1990): *Lean body mass, skin thickness, and lumbar spine density increased, whereas adipose tissue mass decreased.*

Understanding the Discussion. Your primary concerns in the discussion (or conclusion) section are the first and last paragraphs, which typically summarize major findings. The remaining portions are the author’s opportunity to propose mechanisms behind the findings,

recommend future research, and suggest practical applications. But read cautiously; many authors fail to clearly differentiate between study results and discussion speculation. Another important consideration is distinguishing between correlation and causal relationships. Two variables significantly increasing/decreasing means some association between them exists (correlation), not necessarily one caused the other (causal). This distinction is often vague in the discussion. For these reasons, placing greater emphasis on the results rather than the discussion section is crucial.

HGH example (Rudman, 1990): *The discussion’s second sentence immediately cautions readers to not generalize the study results to men with normal HGH levels (which they admit is more than two-thirds of men) or similarly aged women. Warnings continue in the third paragraph; the authors remind readers that HGH replacement*

therapies given to subjects were continually adjusted, not standard-issue pharmacological dosages. Additionally, 2 important points were made when discussing changes in lean body mass and adipose tissue. First, “lean body mass” does not equal skeletal muscle mass but rather is an estimate of all non-fat tissue and includes muscle, skin, liver, kidneys, spleen, and bone. Second, previous animal studies showed HGH supplementation enlarges visceral organ size, leading Rudman et al. (4) to conclude that the increase in lean body mass was partially a result of internal organ enlargement. Furthermore, only lumbar bone density improved, whereas no other bone (jaw, arm, hip, leg, etc) responded to therapy, guiding investigators to conclude that HGH minimally influenced bone health.

STEP 5: DRAW PERSONAL CONCLUSIONS

Finally, after finding and analyzing the article, form personal conclusions based on your interpretation. Start by asking yourself the following 3 questions:

1. Do I agree with the article’s conclusions?

Do not feel forced to accept conclusions of the authors. After scrutinizing the results, how do you interpret the study findings?

HGH example (Rudman, 1990): *No. The authors thankfully acknowledge the study limitations and caution against potential misinterpretations. Yet the abstract states “diminished secretion of HGH is responsible in part for the decrease in lean body mass, expansion of adipose tissue, and the thinning of skin that occurs in old age.” This interpretation is incorrect because the study design does not allow a causal relationship to be drawn. This conclusion emphasizes the danger of only reading abstracts. Reading the discussion section exposes the study limitations and provides a proper interpretation, whereas the abstract’s wording entices readers to draw inappropriate conclusions such as “HGH reverses aging,” “HGH decreases body fat,” or “HGH improves muscle mass.” After reading the entire article, a more appropriate conclusion might be: “six months of continual HGH replacement injections may increase lean body mass, skin and lumbar thickness, and*

decrease adipose tissue mass in HGH diminished 61- to 81-year-old sedentary men. However, much work is still needed on the long term implications of HGH treatment therapy.”

2. Do I agree with conclusions from second-hand sources regarding this article?

Second-hand sources may blatantly falsify, accidentally misinterpret, or erroneously apply conclusions of peer-reviewed data. Consider these claims and determine if they are supported by the evidence you just analyzed.

HGH example (Rudman, 1990): *No. It is important to recognize the authors of the Rudman article have no control over how their results are interpreted by readers. For example, this study is often inappropriately referenced in advertisements as proof that HGH improves sleep, hair quality, energy levels, muscle tone, wrinkle frequency, immune function, and acts as an antiaging drug. Based on our analysis, these statements are complete fallacies because none of these variables were measured in the study. Moreover, skeletal muscle mass was not specifically assessed, let alone “proven” to increase.*

3. Does this directly apply to my athlete/client?

Do the study’s subjects share similar characteristics as your athletes (gender, age, training history, etc)? If not, be careful when applying the results and conclusions to your athletes. Sometimes finding exact matches is not possible. In this case, applying results from studies with comparable subjects is okay.

HGH example (Rudman, 1990): *These results apply only if your athlete is a 61- to 81-year-old man with below-normal HGH levels. It is extremely important to remember that adjusting HGH concentrations from suboptimal to normal has far different physiological consequences than increasing HGH from normal to high concentrations, a moniker true for most mechanisms in the body. Assuming similar responses from a young or middle-aged active/highly trained person with normal/high levels of HGH is illogical.*

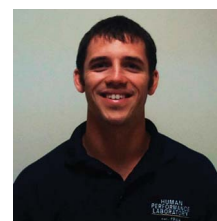
SUMMARY

Expecting coaches to habitually scrutinize all peer-reviewed human

performance research is impractical and unrealistic. Yet paradoxically, athletes rely almost exclusively on coaches, not researchers, to answer various training, diet, and supplement questions. Coaches must therefore transition from passive consumers to critical evaluators by developing skills necessary to find, analyze, and interpret peer-reviewed research. We developed the RAPID method specifically for this purpose, to reinforce the NSCA’s goal of bringing science to application. After applying the RAPID method to the paper of Rudman et al. (4) (see Table 1 for sample), only a minimal relationship was found between HGH supplementation and improved body composition and bone health. Furthermore, the aforementioned review articles make it clear that in 20+ years of research, HGH supplementation has yet to produce significant enhancements of skeletal muscle size or performance in young healthy subjects (3,6). In conclusion, although HGH supplementation may benefit some clinical populations, it does not appear beneficial when abused illegally (3,7).

ACKNOWLEDGMENTS

The authors would like to thank Christopher A. Moore and Mandy M. Ross for their assistance in this project.



Andrew J. Galpin is Assistant Professor in the Center for Sport Performance in the Department of Kinesiology at

California State University, Fullerton.



James R. Bagley is a Doctoral Student in Human Bioenergetics at Ball State University.

REFERENCES

1. Drazen JM. Inappropriate advertising of dietary supplements. *N Engl J Med* 348: 777–778, 2003.
2. Melrose D, Ocker L, Bonnette R, and Spaniol F. The Internet and steroids: A less than honest relationship. *J Strength Cond* 33: 95–99, 2011.
3. Rennie MJ. Claims for the anabolic effects of growth hormone: a case of the emperor's new clothes? *Br J Sports Med* 37: 100–105, 2003.
4. Rudman D, Feller AG, Nagraj HS, Gergans GA, Lalitha PY, Goldberg AF, Schlenker RA, Cohn L, Rudman IW, and Mattson DE. Effects of human growth hormone in men over 60 years old. *N Engl J Med* 323: 1–6, 1990.
5. Vance ML. Growth hormone for the elderly? *N Engl J Med* 323: 52–54, 1990.
6. West DW and Phillips SM. Anabolic processes in human skeletal muscle: restoring the identities of growth hormone and testosterone. *Phys Sportsmed* 38: 97–104, 2010.
7. Yarasheski KE. Growth hormone effects on metabolism, body composition, muscle mass, and strength. *Exerc Sport Sci Rev* 22: 285–312, 1994.



CALL FOR AUTHORS

Have your article published by the NSCA. Share your expertise and reach an audience of over 33,000 subscribers.

Submit your article to NSCA's
Performance Training Journal.

For writer's guidelines, visit...

[http://www.nsca-lift.org/Perform/
PTJ_Writers_Guide.pdf](http://www.nsca-lift.org/Perform/PTJ_Writers_Guide.pdf)